Multi-view Pyramid Transformer: Look Coarser to See Broader

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Abstract

Feed-forward 3D reconstruction from posed images is now fast and accurate for a few views, but scaling the same models to tens or hundreds of high-resolution frames remains difficult. Full multi-view attention becomes prohibitively expensive, while aggressively local or linear approximations often weaken long-range geometric reasoning. We propose Multi-view Pyramid Transformer (MVP), a hierarchical multi-view encoder that expands inter-view context only after progressively reducing intra-view token resolution. MVP augments RGB images with Plucker ray maps and register tokens, applies frame-wise attention at full resolution, alternates frame-wise and group-wise attention after the first token reduction, and alternates frame-wise and global attention after a second reduction. A pyramidal feature aggregation head fuses all three stages and decodes per-pixel 3D Gaussian primitives for direct feed-forward rendering. On DL3DV, MVP consistently outperforms prior feed-forward baselines from 16 to 256 views, improving PSNR from 24.30 to 25.96 at 32 views and from 22.98 to 29.02 at 128 views over iLRM, while remaining sub-second up to 128 views. At 256 views, MVP reaches 29.67 PSNR in 1.84 seconds, while Long-LRM runs out of memory and iLRM degrades to 20.63 PSNR. MVP also generalizes strongly in zero-shot evaluation on Tanks&Temples and Mip-NeRF360, and achieves 33.40 PSNR on 8-view RE10K with a higher-fidelity variant.

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1. Introduction

Feed-forward novel view synthesis and reconstruction models have become an attractive alternative to per-scene optimization because they can infer explicit 3D scene representations in a single forward pass. Recent methods predict radiance fields or 3D Gaussians directly from posed images and drastically reduce reconstruction latency relative to optimization-based systems such as 3D Gaussian Splatting (3D-GS) [16]. This progress is especially visible in large reconstruction models based on transformers and explicit Gaussian decoders [13, 25, 30, 34, 36].

The next bottleneck is scale. Scene-level reconstruction benefits from seeing many views at high resolution, but concatenating all image tokens quickly produces extremely long sequences. A naive solution is to keep global attention everywhere, which scales quadratically with the number of view tokens [26]. Prior work therefore resorts to linear attention [15], sparse or windowed patterns [3, 9, 32], local view neighborhoods [12], or compressed scene tokens [29]. These strategies improve efficiency, but they often trade away either broad view-to-view communication or fine-grained spatial evidence. Empirically, this tension is visible in existing scene-scale feed-forward models: performance may improve initially as more views are added, but it can also saturate, collapse, or exhaust memory when the input context becomes substantially longer [13, 36].

Figure 1. MVP reconstructs scenes across a wide range of view counts while preserving a favorable quality-speed trade-off.

Our key observation is that inter-view context and intra-view resolution should not be scaled independently. Broad global reasoning is most useful after each view has already distilled its local evidence, while fine-resolution tokens are most useful before attention spans many views. Based on this observation, we propose *Multi-view Pyramid Transformer* (MVP), a dual hierarchy that expands view context from frame to group to global while compressing image tokens from fine to coarse. MVP starts with full-resolution frame-wise reasoning, then merges neighboring spatial tokens, performs alternating frame-wise and group-wise attention within small view groups, reduces tokens again, and finally performs alternating frame-wise and global attention on a much shorter sequence. The architecture preserves local detail early, introduces an intermediate group level that bridges frame-only and fully global reasoning, and confines

the most expensive global interactions to the coarsest token stage.

The released implementation also reveals two practical ingredients that make this hierarchy effective. First, each image is augmented with Plucker ray maps and processed with register tokens [10], while group/global attention uses camera-aware relative positional encoding via PRoPE [18]. Second, MVP does not rely only on the final coarse stage: a pyramidal feature aggregation head, inspired by dense prediction transformers and feature pyramids [19, 23, 24], fuses features from all three stages before decoding dense per-pixel 3D Gaussian parameters.

Experiments support this design. On DL3DV [20], MVP outperforms prior feed-forward baselines from 16 to 256 input views and remains fast as the context grows. On the challenging 256-view setting, it reaches 29.67 PSNR in 1.84 seconds, whereas Long-LRM [36] runs out of memory and iLRM [13] drops to 20.63 PSNR in 20.92 seconds. MVP also improves zero-shot transfer to Tanks&Temples [17] and Mip-NeRF360 [2], and achieves strong RE10K [35] results with both coarse and fine variants.

Our contributions are threefold. First, we introduce a dual-hierarchy transformer for scalable multi-view reconstruction that couples local-to-global inter-view reasoning with fine-to-coarse intra-view token reduction. Second, we show that group-wise attention is a useful intermediate regime between frame-only and fully global attention, improving both reconstruction quality and scaling behavior. Third, we demonstrate that pyramidal multi-stage feature aggregation is important for dense 3D Gaussian prediction, yielding a strong accuracy-speed trade-off across in-domain, longer-context, and zero-shot settings.

2. Related Work

Feed-forward 3D reconstruction and Gaussian prediction. Recent reconstruction models replace per-scene optimization with feed-forward inference, often decoding explicit 3D representations such as Gaussian splats. Early large reconstruction models focused on sparse-view or object-centric settings [5, 8, 25, 30, 34]. More recent scene-level methods extend this idea to wider baselines and larger view counts [6, 12, 13, 29, 36]. LaRa [6] already argues that purely global attention is not ideal for reconstruction and introduces grouped reasoning, but it does not build a multi-stage spatial pyramid for long, high-resolution contexts. Long-LRM [36] handles high-resolution sequences through token merging and hybrid blocks, yet still uses a long-sequence design centered around a flat token stream. iLRM [13] improves efficiency by iterative refinement and disentangling scene and image representations, but its performance degrades sharply as the number of views grows in our evaluation. MVP differs by explicitly coupling inter-view hierarchy and intra-view token reduction inside one feed-forward encoder.

Multi-view and general 3D vision transformers. Transformers have become a common backbone for geometric prediction, from DUSt3R [28] and VGGT [27] to pose-free Gaussian reconstruction [14] and higher-fidelity Gaussian densification [22]. These models confirm that large multi-view models can learn strong geometric priors from data, but they do not directly solve the efficiency problem posed by dense, long multi-view input sequences for scene rendering. MVP targets this particular regime: posed multi-view reconstruction with tens to hundreds of input images and dense per-pixel Gaussian prediction.

Efficient attention and hierarchical vision models. Long-context architectures often reduce the quadratic cost of attention through linearization [15], sparse patterns [9, 32], or local-global hybrids [3]. In vision, hierarchical token pyramids and divided attention are also well established [1, 4, 21]. MVP is related in spirit, but its hierarchy is designed for posed multi-view reconstruction: frame-wise attention extracts within-view evidence, group-wise attention connects moderately sized neighborhoods of views, and global attention is deferred until token counts are much smaller. We further combine ray-map tokenization with register tokens [10] and camera-aware PRoPE attention [18], which makes the grouped and global interactions geometry-aware.

3. Method

We study feed-forward reconstruction from V posed RGB images $\{I_v, K_v, T_v\}_{v=1}^{V}$, where K_v and T_v denote camera intrinsics and extrinsics. The goal is to predict a set of 3D Gaussian primitives that can be rasterized into target views without test-time optimization.

3.1. Ray-aware tokenization

For each input image, MVP concatenates RGB with a Plucker ray map built from the ray origin $\mathbf{o}$, ray direction $\mathbf{d}$, and $\mathbf{o} \times \mathbf{d}$. In the released code this yields 12 input channels per pixel: 3 for RGB and 9 for geometry. Images are patchified with patch size 8 and projected to a stage-1 embedding dimension of 256. We also prepend four learned register tokens per view [10], which provide persistent workspace tokens during attention.

This representation grounds each token in both appearance and camera geometry from the first layer onward. In contrast to using image features alone, the model does not need to infer viewing rays implicitly.

3.2. Dual hierarchy: frame, group, and global attention

Figure 2 summarizes MVP. The encoder has three stages with widths 256, 512, and 1024 and depths 2, 4, and 8, respectively. The key design is to expand inter-view context only after reducing intra-view token count.

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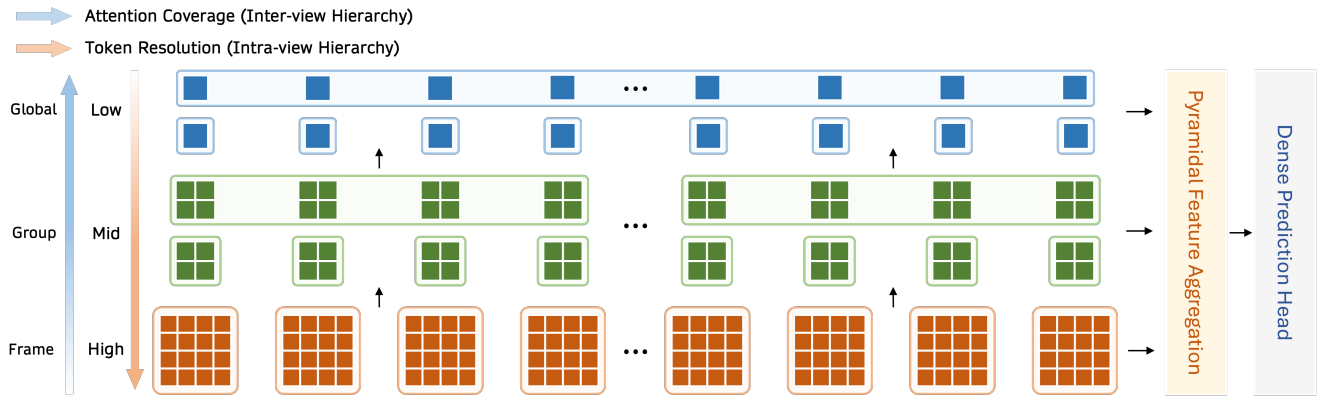

Figure 2. Overview of MVP. Posed images are tokenized from RGB and Plucker ray maps, processed by a three-stage pyramid with frame-wise, group-wise, and global interactions, fused by a pyramidal aggregation head, and decoded into dense 3D Gaussian primitives for feed-forward rendering. MVP broadens inter-view communication only after reducing spatial token count.

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Stage 1: frame-wise attention. Stage 1 operates independently on each view. Tokens from one image and its register tokens are processed by standard self-attention and MLP blocks. This stage extracts view-local evidence at full spatial resolution, where precise geometric correspondences are most spatially anchored.

Token reduction. After Stage 1, image tokens are reshaped back to a 2D grid and passed through a stride-2 depthwise convolution that halves height and width while doubling channel width. Register tokens are projected with a linear layer to the new width. The same operation is repeated between Stages 2 and 3. After two reductions, each view has only one sixteenth as many spatial tokens as at the input stage.

Stage 2: alternating frame-wise and group-wise attention. Views are partitioned into groups of four in the released model. Stage 2 alternates between frame-wise blocks and group-wise blocks. The frame-wise blocks refine each view independently at the reduced spatial resolution. The group-wise blocks concatenate the views within a group and apply geometry-aware attention with PRoPE [18]. This intermediate regime is central to MVP: it shares information across nearby views without paying the cost of full global attention across all views.

Stage 3: alternating frame-wise and global attention. After the second token reduction, Stage 3 again alternates frame-wise and cross-view blocks, but now the cross-view blocks attend over all input views. Because this stage sees coarsened spatial tokens, it can reason globally over much longer contexts than a flat high-resolution transformer. The alternating pattern retains within-view refinement even at the coarsest level.

The hierarchy can be understood as a deliberate scheduling of attention bandwidth. If L is the number of stage-1 spatial tokens per view, then Stage 2 attends over approximately $L/4$ tokens per view and Stage 3 over $L/16$. The most expensive global attention is therefore applied only after a $16\times$ reduction in per-view token count, while Stage 2 scales linearly with the number of groups for fixed group size. This is the core mechanism behind the quality-speed trade-off in MVP.

3.3. Pyramidal feature aggregation

Using only the coarsest stage would discard fine details. MVP therefore fuses features from all three stages using a pyramidal aggregation head based on the DPT-style dense prediction design [23] with multi-scale fusion blocks reminiscent of FPN/U-Net [19, 24]. Stage-1, Stage-2, and Stage-3 tokens are reshaped into feature maps, projected to a common feature width, and fused top-down to recover a dense output grid. The ablations in Table 5 show that this module is not cosmetic: removing it reduces PSNR from 22.79 to 21.58 and degrades SSIM and LPIPS substantially.

3.4. Dense Gaussian prediction

The fused dense tokens are decoded into one Gaussian prediction per input pixel. The released model predicts a ray-aligned parameterization consisting of 3D center parameters, spherical-harmonic appearance coefficients, anisotropic scales, rotation, and view-dependent opacity coefficients. Concretely, the implementation outputs:

- 3 values used to derive a distance-like quantity along the input ray,
- first-order spherical harmonics for color,
- 3 log-scales,
- a 4D rotation representation,
- second-order spherical harmonics for opacity.

The Gaussian center is obtained by projecting the predicted distance along the input camera ray, which keeps predictions tied to valid viewing geometry. Target views are rendered with a differentiable Gaussian rasterizer built on top of `gsplat` [31].

3.5. Training

The released training recipe uses a three-stage curriculum. Stage 1 trains a lower-resolution model at roughly 256×480 with 32 input views. Stage 2 increases the resolution to roughly 544×960. A final training stage freezes earlier components and varies the number of input views from 16 to 128. The optimization objective combines pixel-wise MSE, a VGG-based perceptual loss, and a small opacity regularizer. Attention uses query-key normalization and RMS-style head normalization in the implementation [11, 33], and training uses gradient checkpointing for memory efficiency [7].

4. Experiments

We first evaluate on DL3DV [20], the main large-scale scene dataset used by recent scene-level feed-forward models. We then test zero-shot generalization on Tanks&Temples [17] and Mip-NeRF360 [2], and report additional results on RE10K [35]. Across all experiments we report PSNR, SSIM, LPIPS, and inference time when available. We compare primarily against 3D-GS$_{30k}$ [16], Long-LRM [36], iLRM [13], and CLiFT [29]. For the 16-view DL3DV setting, Table 1 follows the provided evaluation protocol and reports re-evaluated 32-view checkpoints for Long-LRM and iLRM, alongside the values reported in their original papers.

4.1. Main results on DL3DV

Tables 1 and 2 show that MVP improves consistently over prior feed-forward baselines as the number of views grows. At 32 views, MVP reaches 25.96 PSNR, exceeding Long-LRM by 1.99 dB and iLRM by 1.66 dB while also being 4.9$\times$ and 3.1$\times$ faster, respectively. At 128 views, the gap widens further: MVP obtains 29.02 PSNR, compared with 21.24 for Long-LRM and 22.98 for iLRM, while remaining below one second. At 192 and 256 views, Long-LRM runs out of memory and iLRM deteriorates severely, whereas MVP remains stable at 29.54 and 29.67 PSNR in 1.23 and 1.84 seconds.

The comparison to optimization-based 3D-GS is also informative. MVP is not uniformly better in absolute quality at very high view counts: 3D-GS remains ahead by 0.41 dB at 128 views and 0.72 dB at 256 views. However, it requires about eight minutes per scene, whereas MVP reconstructs the same settings in well under two seconds. At 32 and 64 views, MVP even exceeds 3D-GS in PSNR, showing that the feed-forward prior is especially helpful in more ambiguous sparse-to-medium view regimes.

The scaling trend is equally important. Long-LRM and iLRM remain competitive around 32 views, but both lose accuracy once cross-view sequences become much longer, whereas MVP continues to improve. This is the regime targeted by the pyramid: all-view communication is deferred until the spatial token budget has already been compressed, so additional views contribute more geometry than overhead.

4.2. Zero-shot transfer and RE10K

Table 3 shows strong zero-shot transfer. On Tanks&Temples, MVP improves from 19.22 to 22.36 PSNR over iLRM in the 128-view setting; on Mip-NeRF360 the gain is from 21.32 to 25.12 PSNR. The improvement grows with more views, suggesting that the hierarchical context expansion generalizes beyond the DL3DV training domain rather than overfitting to a specific camera pattern.

This pattern is consistent with the role of the group stage. By aggregating evidence over moderate view neighborhoods before full-scene mixing, MVP learns correspondences that are less tied to one dataset's camera spacing and therefore transfer better to new trajectories and scene scales.

Table 4 reports RE10K results. The coarse MVP variant already improves over CLiFT and iLRM in both the 4-view and 8-view settings, and the fine variant reaches 32.12 PSNR with 4 views and 33.40 PSNR with 8 views. Figure 4 shows that the fine variant especially reduces residual errors around high-contrast window frames and foliage.

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Method	16 views				32 views				64 views				128 views			
	PSNR ↑	SSIM ↑	LPIPS ↓	Time (s) ↓	PSNR ↑	SSIM ↑	LPIPS ↓	Time (s) ↓	PSNR ↑	SSIM ↑	LPIPS ↓	Time (s) ↓	PSNR ↑	SSIM ↑	LPIPS ↓	Time (s) ↓
3D-GS$_{30k}$ [16]	21.48	0.753	0.252	8min	24.43	0.827	0.191	8min	27.34	0.883	0.146	8min	29.43	0.914	0.123	8min
Long-LRM [36]	21.05 (22.66)	0.708 (0.740)	0.297 (0.292)	0.50	23.97	0.778	0.267	0.84	23.60	0.789	0.260	2.08	21.24	0.739	0.308	6.39
iLRM [13]	21.92 (22.91)	0.748 (0.766)	0.316 (0.295)	0.19	24.30	0.803	0.256	0.53	24.44	0.819	0.240	1.66	22.98	0.807	0.249	5.61
Ours	**23.76**	**0.798**	**0.239**	**0.09**	**25.96**	**0.847**	**0.187**	**0.17**	**27.73**	**0.881**	**0.154**	**0.36**	**29.02**	**0.903**	**0.134**	**0.77**

Table 1. Quantitative comparisons on the DL3DV dataset with varying numbers of input views. The values in parentheses $(\cdot)$, for the 16-view setting are taken from the original papers, where the models are trained with a fixed 16-view input. For all metrics, we report the results by re-evaluating the models from their 32-view checkpoints.

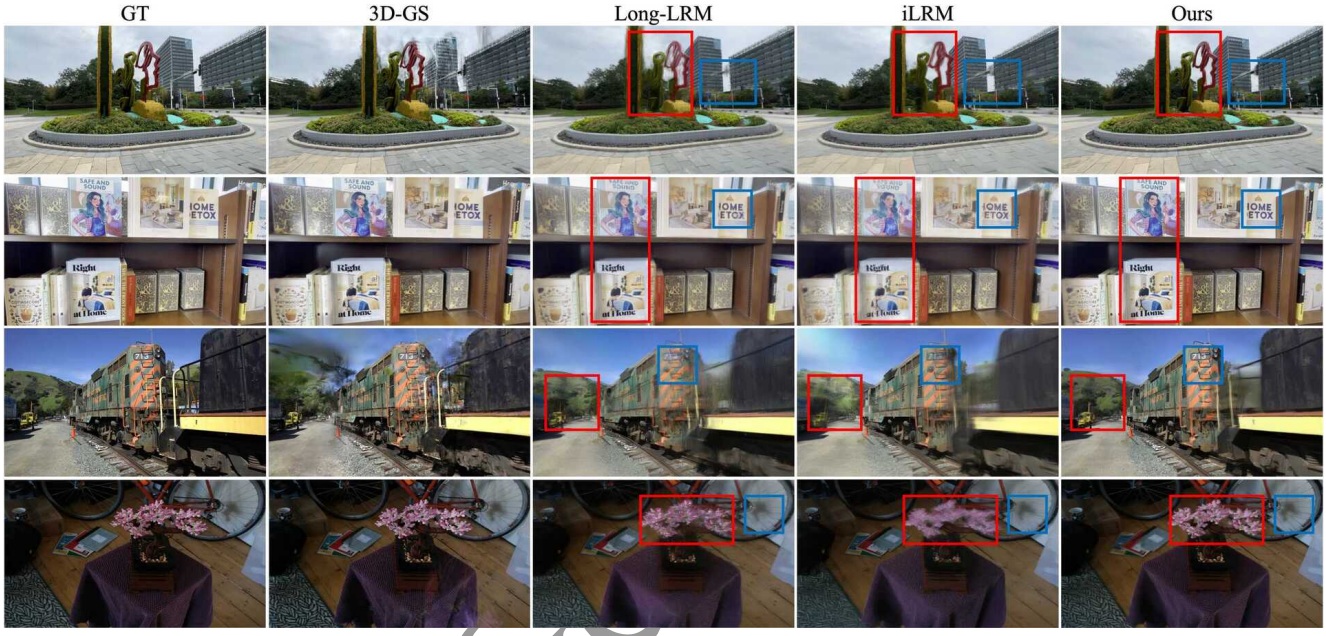

Figure 3. Qualitative comparison on DL3DV, Tanks&Temples, and Mip-NeRF360. MVP produces sharper structures and fewer cross-view inconsistencies than Long-LRM and iLRM, especially on thin objects, text, and distant scene content.

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Method	192 views				256 views			
	PSNR ↑	SSIM ↑	LPIPS ↓	Time (s) ↓	PSNR ↑	SSIM ↑	LPIPS ↓	Time (s) ↓
3D-GS$_{30k}$ [16]	30.14	0.924	0.116	8min	30.39	0.926	0.114	8min
Long-LRM [36]	OOM (GPU memory limit exceeded, 80 GB)							
iLRM [13]	21.68	0.788	0.264	11.91	20.63	0.767	0.281	20.92
Ours	**29.54**	**0.912**	**0.127**	**1.23**	**29.67**	**0.915**	**0.128**	**1.84**

Table 2. Quantitative comparisons on the DL3DV dataset with varying numbers of input views.

Method	Views	Tanks & Temples			Mip-NeRF360		
		PSNR ↑	SSIM ↑	LPIPS ↓	PSNR ↑	SSIM ↑	LPIPS ↓
Long-LRM [36]	32	18.59	0.614	0.366	21.08	0.484	0.445
iLRM [13]		18.58	0.631	0.385	21.09	0.495	0.466
Ours		**19.54**	**0.708**	**0.277**	**22.21**	**0.587**	**0.355**
Long-LRM	64	19.44	0.651	0.334	21.30	0.499	0.431
iLRM		19.82	0.692	0.318	21.60	0.522	0.444
Ours		**21.24**	**0.761**	**0.221**	**23.72**	**0.656**	**0.302**
Long-LRM	128	18.47	0.613	0.375	19.82	0.484	0.457
iLRM		19.22	0.696	0.319	21.32	0.551	0.424
Ours		**22.36**	**0.804**	**0.184**	**25.12**	**0.736**	**0.248**

Table 3. Quantitative comparisons on the Tanks&Temples and Mip-NeRF360 under a zero-shot setting. For the 128 view setting on Mip-NeRF360, the `stump` and `treehill` scenes are excluded due to their limited number of frames.

5. Ablation and Analysis

5.1. Why does the hierarchy help?

Table 5 isolates the main design choices on DL3DV at reduced resolution. Several conclusions are immediate. First, pyramidal feature aggregation is essential for dense prediction quality: removing it causes a large drop in PSNR, SSIM, and LPIPS. Second, group-wise attention is not replaceable by either more frame-wise blocks or more global blocks. Replacing the group stage with frame-only reasoning lowers PSNR to 22.53, while replacing it with earlier global reasoning gives 22.94, both below the 22.79 baseline. Third, the

Method	4 views			8 views		
	PSNR ↑	SSIM ↑	LPIPS ↓	PSNR ↑	SSIM ↑	LPIPS ↓
CLiFT [29]	30.13	0.916	0.095	29.68	0.910	0.092
iLRM [13]	30.37	0.923	0.095	28.90 (31.57)	0.920 (0.935)	0.102 (0.082)
Ours-coarse	30.56	0.925	0.087	31.78	0.939	0.077
Ours-fine	**32.12**	**0.941**	**0.077**	**33.40**	**0.952**	**0.066**

Table 4. Quantitative comparisons on the RE10K dataset. Baselines are evaluated using 4-view models, and values in parentheses $(\cdot)$ are from the models trained for that specific view setting.

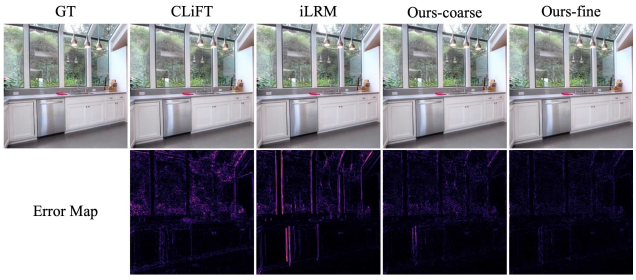

Figure 4. RE10K qualitative comparison. The fine MVP variant yields the smallest error map and the cleanest recovery of thin structures and reflective boundaries.

full dual hierarchy matters more than either hierarchy alone. Removing the inter-view hierarchy, removing the intra-view hierarchy, or collapsing both into flat global models all hurt quality, and reversing the hierarchy is catastrophic, reducing PSNR to 18.95.

These ablations also support the paper's central intuition: broad global context is most effective after local evidence has been compressed into stronger coarse tokens. Applying global reasoning too early is not only slower but also less accurate.

5.2. Longer-context generalization

Table 6 evaluates a model trained with 32 input views on unseen 40-view and 48-view settings. MVP not only remains stable; it improves meaningfully as more views are provided, rising from 25.88 to 26.36 and then 27.06 PSNR, while inference time increases only modestly from 0.17 to 0.26 seconds. The baseline models also benefit from more views, but the gains are much smaller. This result is important because it shows that MVP does not merely memorize one training context length. The hierarchy transfers to longer sequences without retraining.

The latency trend reinforces the same point: moving from 32 to 48 views adds only 0.09 seconds for MVP, versus 0.54 for Long-LRM and 0.51 for iLRM. The extra quality therefore does not come from spending comparable extra compute at test time.

5.3. Attention behavior

Figure 6 visualizes attention patterns in Stage 2 and Stage 3. In the group-wise stage, high-attention regions remain concentrated within a small set of neighboring views, which is appropriate for establishing medium-range correspondences. In the global stage, attention expands to semantically corresponding regions across distant viewpoints. The visualization aligns with the design intent of MVP: group attention acts as a bridge that prevents the model from jumping directly from isolated frame reasoning to diffuse full-context attention.

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Method	Stage 1	Stage 2	Stage 3	Patch Size	Hidden Dim.	Feature Agg.	PSNR ↑	SSIM ↑	LPIPS ↓
Baseline	Frame(2)	Group(4)	Global(8)	(8, 16, 32)	(128, 256, 512)	o	22.79	0.733	0.235
w/o Feature Agg.	Frame(2)	Group(4)	Global(8)	(8, 16, 32)	(128, 256, 512)	x	21.58	0.646	0.340
w/o Group-attn.	Frame(2)	Frame(4)	Global(8)	(8, 16, 32)	(128, 256, 512)	o	22.53	0.720	0.247
	Frame(2)	Global(4)	Global(8)				22.94	0.739	0.235
w/o Inter-view Hierarchy	Global(2)	Global(4)	Global(8)	(8, 16, 32)	(128, 256, 512)	o	22.94	0.739	0.236
w/o Intra-view Hierarchy	Frame(2)	Group(4)	Global(8)	(8, 8, 8)	(512, 512, 512)	o	22.83	0.732	0.249
w/o Dual Hierarchy (p=8)	Global(2)	Global(4)	Global(8)	(8, 8, 8)	(512, 512, 512)	o	23.20	0.747	0.241
w/o Dual Hierarchy (p=16)	Global(2)	Global(4)	Global(8)	(16, 16, 16)	(512, 512, 512)	o	21.80	0.651	0.341
Reversed Hierarchy	Global(8)	Group(4)	Frame(2)	(32, 16, 8)	(512, 256, 128)	o	18.95	0.442	0.555

Table 5. Ablations. We train all variants for 100K iterations at a reduced resolution (256×256) on the DL3DV dataset.

Method	32 views (Trained)				40 views (Unseen)				48 views (Unseen)			
	PSNR ↑	SSIM ↑	LPIPS ↓	Time (s) ↓	PSNR ↑	SSIM ↑	LPIPS ↓	Time (s) ↓	PSNR ↑	SSIM ↑	LPIPS ↓	Time (s) ↓
Long-LRM [36]	23.97	0.778	0.267	0.84	24.18 (0.21)	0.787	0.260	1.05 (0.21)	24.30 (0.33)	0.797	0.252	1.38 (0.54)
iLRM [13]	24.30	0.803	0.257	0.53	24.54 (0.24)	0.811	0.248	0.76 (0.23)	24.78 (0.48)	0.820	0.240	1.04 (0.51)
Ours	**25.88**	**0.845**	**0.188**	**0.17**	**26.36** (0.48)	**0.855**	**0.178**	**0.22** (0.05)	**27.06** (1.18)	**0.870**	**0.164**	**0.26** (0.09)

Table 6. Longer context generalization. Values in parentheses $(\cdot)$ denote the improvements over 32 views.

6. Conclusion

MVP addresses a concrete scaling problem in feed-forward multi-view reconstruction: how to use many high-resolution posed images without paying the full cost of global attention on every token. The central idea is to broaden view-to-view communication only after coarsening spatial tokens. This yields a simple but effective dual hierarchy of frame-wise, group-wise, and global reasoning, complemented by multi-scale feature aggregation for dense Gaussian decoding. Across DL3DV, RE10K, Tanks&Temples, and Mip-NeRF360, the resulting model offers a strong quality-speed trade-off and remains stable as the number of input views increases. The ablations and attention visualizations indicate that the intermediate group stage and the fine-to-coarse spatial pyramid are both necessary parts of that behavior.

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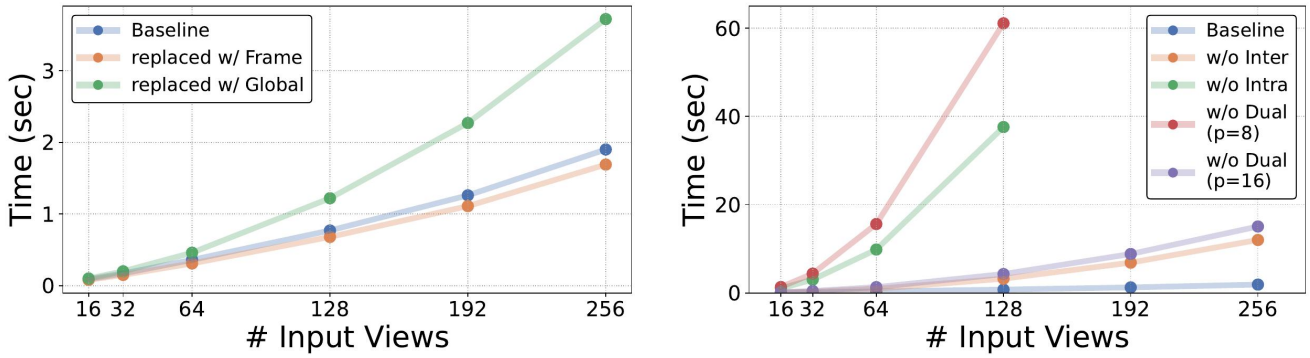

Figure 5. Inference-time scaling of the ablations. Left: replacing group-wise attention with full global attention increases latency much faster than the baseline, whereas replacing it with frame-only attention is cheaper but less expressive. Right: removing the intra-view pyramid, especially in the flat global setting, causes inference time to grow sharply as the number of views increases.

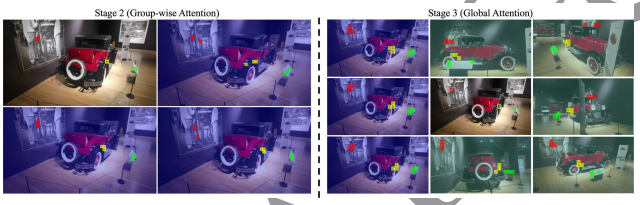

Figure 6. Attention visualization. Group-wise attention focuses within local view groups, while the later global stage links corresponding scene regions across distant cameras.

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References

[1] Anurag Arnab, Mostafa Dehghani, Georg Heigold, Chen Sun, Mario Lučić, and Cordelia Schmid. Vivit: A video vision transformer. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 6836–6846, 2021.

[2] Jonathan T. Barron, Ben Mildenhall, Dor Verbin, Pratul P. Srinivasan, and Peter Hedman. Mip-nerf 360: Unbounded anti-aliased neural radiance fields. *CVPR*, 2022.

[3] Iz Beltagy, Matthew E Peters, and Arman Cohan. Longformer: The long-document transformer. *arXiv preprint arXiv:2004.05150*, 2020.

[4] Gedas Bertasius, Heng Wang, and Lorenzo Torresani. Is space-time attention all you need for video understanding? In *Icml*, page 4, 2021.

[5] David Charatan, Sizhe Lester Li, Andrea Tagliasacchi, and Vincent Sitzmann. pixelsplat: 3d gaussian splats from image pairs for scalable generalizable 3d reconstruction. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 19457–19467, 2024.

[6] Anpei Chen, Haofei Xu, Stefano Esposito, Siyu Tang, and Andreas Geiger. Lara: Efficient large-baseline radiance fields. In *European Conference on Computer Vision*, pages 338–355. Springer, 2024.

[7] Tianqi Chen, Bing Xu, Chiyuan Zhang, and Carlos Guestrin. Training deep nets with sublinear memory cost. *arXiv preprint arXiv:1604.06174*, 2016.

[8] Yuedong Chen, Haofei Xu, Chuanxia Zheng, Bohan Zhuang, Marc Pollefeys, Andreas Geiger, Tat-Jen Cham, and Jianfei Cai. Mvsplat: Efficient 3d gaussian splatting from sparse multi-view images. In *European Conference on Computer Vision*, pages 370–386. Springer, 2024.

[9] Rewon Child, Scott Gray, Alec Radford, and Ilya Sutskever. Generating long sequences with sparse transformers. *arXiv preprint arXiv:1904.10509*, 2019.

[10] Timothée Darcet, Maxime Oquab, Julien Mairal, and Piotr Bojanowski. Vision transformers need registers. *arXiv preprint arXiv:2309.16588*, 2023.

[11] Alex Henry, Prudhvi Raj Dachapally, Shubham Pawar, and Yuxuan Chen. Query-key normalization for transformers. *arXiv preprint arXiv:2010.04245*, 2020.

[12] Tooba Imtiaz, Lucy Chai, Kathryn Heal, Xuan Luo, Jungyeon Park, Jennifer Dy, and John Flynn. Lvt: Large-scale scene reconstruction via local view transformers. *arXiv preprint arXiv:2509.25001*, 2025.

[13] Gyeongjin Kang, Seungtae Nam, Seungkwon Yang, Xiangyu Sun, Sameh Khamis, Abdelrahman Mohamed, and Eunbyung Park. ilrm: An iterative large 3d reconstruction model. *arXiv preprint arXiv:2507.23277*, 2025.

[14] Gyeongjin Kang, Jisang Yoo, Jihyeon Park, Seungtae Nam, Hyeonsoo Im, Sangheon Shin, Sangpil Kim, and Eunbyung Park. Selfsplat: Pose-free and 3d prior-free generalizable 3d gaussian splatting. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 22012–22022, 2025.

[15] Angelos Katharopoulos, Apoorv Vyas, Nikolaos Pappas, and François Fleuret. Transformers are rnns: Fast autoregressive transformers with linear attention. In *International conference on machine learning*, pages 5156–5165. PMLR, 2020.

[16] Bernhard Kerbl, Georgios Kopanas, Thomas Leimkühler, and George Drettakis. 3d gaussian splatting for real-time radiance field rendering. *ACM Trans. Graph.*, 42(4):139–1, 2023.

[17] Arno Knapitsch, Jaesik Park, Qian-Yi Zhou, and Vladlen Koltun. Tanks and temples: Benchmarking large-scale scene reconstruction. *ACM Transactions on Graphics*, 36(4), 2017.

[18] Ruilong Li, Brent Yi, Junchen Liu, Hang Gao, Yi Ma, and Angjoo Kanazawa. Cameras as relative positional encoding. *Advances in Neural Information Processing Systems*, 2025.

[19] Tsung-Yi Lin, Piotr Dollár, Ross Girshick, Kaiming He, Bharath Hariharan, and Serge Belongie. Feature pyramid networks for object detection. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 2117–2125, 2017.

[20] Lu Ling, Yichen Sheng, Zhi Tu, Wentian Zhao, Cheng Xin, Kun Wan, Lantao Yu, Qianyu Guo, Zixun Yu, Yawen Lu, et al. Dl3dv-10k: A large-scale scene dataset for deep learning-based 3d vision. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 22160–22169, 2024.

[21] Ze Liu, Yutong Lin, Yue Cao, Han Hu, Yixuan Wei, Zheng Zhang, Stephen Lin, and Baining Guo. Swin transformer: Hierarchical vision transformer using shifted windows. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021.

[22] Seungtae Nam, Xiangyu Sun, Gyeongjin Kang, Younggeun Lee, Seungjun Oh, and Eunbyung Park. Generative densification: Learning to densify gaussians for high-fidelity generalizable 3d reconstruction. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 26683–26693, 2025.

[23] René Ranftl, Alexey Bochkovskiy, and Vladlen Koltun. Vision transformers for dense prediction. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 12179–12188, 2021.

[24] Olaf Ronneberger, Philipp Fischer, and Thomas Brox. U-net: Convolutional networks for biomedical image segmentation. In *International Conference on Medical*

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image computing and computer-assisted intervention, pages 234–241. Springer, 2015.

[25] Jiaxiang Tang, Zhaoxi Chen, Xiaokang Chen, Tengfei Wang, Gang Zeng, and Ziwei Liu. Lgm: Large multi-view gaussian model for high-resolution 3d content creation. In *European Conference on Computer Vision*, pages 1–18. Springer, 2024.

[26] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. *Advances in neural information processing systems*, 30, 2017.

[27] Jianyuan Wang, Minghao Chen, Nikita Karaev, Andrea Vedaldi, Christian Rupprecht, and David Novotny. Vggt: Visual geometry grounded transformer. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 5294–5306, 2025.

[28] Shuzhe Wang, Vincent Leroy, Yohann Cabon, Boris Chidlovskii, and Jerome Revaud. Dust3r: Geometric 3d vision made easy. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 20697–20709, 2024.

[29] Zhengqing Wang, Yuefan Wu, Jiacheng Chen, Fuyang Zhang, and Yasutaka Furukawa. Clift: Compressive light-field tokens for compute-efficient and adaptive neural rendering. *arXiv preprint arXiv:2507.08776*, 2025.

[30] Yinghao Xu, Zifan Shi, Wang Yifan, Hansheng Chen, Ceyuan Yang, Sida Peng, Yujun Shen, and Gordon Wetzstein. Grm: Large gaussian reconstruction model for efficient 3d reconstruction and generation. In *European Conference on Computer Vision*, pages 1–20. Springer, 2024.

[31] Vickie Ye, Ruilong Li, Justin Kerr, Matias Turkulainen, Brent Yi, Zhuoyang Pan, Otto Seiskari, Jianbo Ye, Jeffrey Hu, Matthew Tancik, et al. gsplat: An open-source library for gaussian splatting. *Journal of Machine Learning Research*, 26(34):1–17, 2025.

[32] Manzil Zaheer, Guru Guruganesh, Kumar Avinava Dubey, Joshua Ainslie, Chris Alberti, Santiago Ontanon, Philip Pham, Anirudh Ravula, Qifan Wang, Li Yang, et al. Big bird: Transformers for longer sequences. *Advances in neural information processing systems*, 33:17283–17297, 2020.

[33] Biao Zhang and Rico Sennrich. Root mean square layer normalization. *Advances in Neural Information Processing Systems*, 2019.

[34] Kai Zhang, Sai Bi, Hao Tan, Yuanbo Xiangli, Nanxuan Zhao, Kalyan Sunkavalli, and Zexiang Xu. Gs-lrm: Large reconstruction model for 3d gaussian splatting. In *European Conference on Computer Vision*, pages 1–19. Springer, 2024.

[35] Tinghui Zhou, Richard Tucker, John Flynn, Graham Fyffe, and Noah Snavely. Stereo magnification: Learning view synthesis using multiplane images. *arXiv preprint arXiv:1805.09817*, 2018.

[36] Chen Ziwen, Hao Tan, Kai Zhang, Sai Bi, Fujun Luan, Yicong Hong, Li Fuxin, and Zexiang Xu. Long-lrm: Long-sequence large reconstruction model for wide-coverage gaussian splats. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 4349–4359, 2025.

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